

EXPERIMENT 11

Student Name: Pranav Sood

UID: 24BDA70188

Branch: AIT-CSE

Section/Group: 24AIT_KRG-G2

Semester: 5

Subject Name: ADBMS

Subject Code: 24CSP-333

Experiment 8 -- To study and implement Window Function

Question :8 Part A (i): Write an SQL query to find the employees whose salary has increased in every recorded month compared to the immediately previous month.

SQL Query:

```
Select E.emid, E.emp_name
FROM (
    select *, LAG(salary,1) OVER(partition by emid ORDER BY s_month) as prev_salary
    FROM EmployeeSalaryHistory
) as E
GROUP BY E.emid, E.emp_name
HAVING COUNT(*) FILTER(
    WHERE salary <= prev_salary
) = 0;
```

	employee_id integer	employee_name character varying (50)
1	103	Raj
2	101	Alok

Question: Part A (ii) Write an SQL query to find the employees whose salaries belong to the top three distinct salary levels within their respective departments.

SQL Query:

EXPERIMENT 11

```

SELECT D.name AS Department,
       E.name AS Employee,
       E.salary
FROM (
  SELECT *,
         DENSE_RANK() OVER (
           PARTITION BY departmentId
           ORDER BY salary DESC
         ) AS rnk
  FROM Employee
) E
JOIN Department D
ON E.departmentId = D.id
WHERE rnk <= 3;
  
```

	department character varying (50) 🔒	employee character varying (50) 🔒	salary integer 🔒
1	IT	Max	90000
2	IT	Jim	85000
3	IT	John	85000
4	IT	Alex	75000
5	Sales	Henry	80000
6	Sales	Susan	75000
7	Sales	David	70000
8	HR	Tina	60000
9	HR	Rita	55000
10	HR	Manoj	50000

Learning Outcome

After completing this experiment, the learner will be able to:

1. Understand and implement Windows Functions

EXPERIMENT 11

2. **Apply aggregate functions** such as Count(), Filter()
3. **Use joins and filtering techniques** to combine data from multiple tables and generate meaningful reports.